

Customer Segmentation and Recommendation System For an Online Retailing Business

Problem Statement:

Considering an online retail store, how can we segment customers into different groups and then recommend products based on a historical transaction dataset?

Context:

To expand business, an online retailing company wants to have a comprehensive analysis of the customers and products. Therefore, the company wants to segment customers based on shopping behavior and then recommend them most related products.

Criteria for Success:

Divide customers into optimal segments and offer most related products using machine learning techniques.

Scope of Solution Space:

Using machine learning techniques, and based on a historical transaction database, we want to group customers based on shopping trends. After customer segmentation, we need to find most related products for each segment. As the result of this segmentation, customers will be receiving related promotions and advertisements. A comprehensive data analysis with data visualization will be sent to stakeholders as the final report.

Constraints:

- 1) Data source includes transactions for only two years, 2010 and 2011.
- 2) The currency is not mentioned, but I assume it is in US Dollars.

Stakeholders:

The online retail company marketing team, executive sales manager, and the IT Development team are stakeholders for this project.

Data Sources:

The data is captured from [Kaggle.com](https://www.kaggle.com), but the original source is UCI Machine Learning Repository. The dataset is in the format of an Excel file and includes transactions for the year 2010 and 2011. Here is a link to access this data source:

<https://www.kaggle.com/code/farzadnekouei/customer-segmentation-recommendation-system>